Arduino Stella User Manual

Learn about the hardware and software features of the Arduino Stella.

Author • José Bagur Last revision • 09/03/2025

This user manual provides a comprehensive overview of the Arduino Stella, highlighting its hardware and software elements. With it, you will learn how to set up, configure and use all the main features of the Arduino Stella.

Hardware and Software Requirements

Hardware Requirements

- Arduino Stella (SKU: ABX00131) (x1)
- Portenta UWB Shield (SKU: ASX00074) (x1)
- Portenta C33 (SKU: ABX00074) (x1)
- USB-C® cable (SKU: TPX00094) (x2)
- CR2032 battery (x1) (optional)

Software Requirements

- Arduino IDE 2.0+
- `StellaUWB` library (designed for the Arduino Stella)
- `PortentaUWBShield` library (designed for the Portenta UWB Shield)
- `ArduinoBLE` library
- Arduino Mbed OS Stella Boards core (required for the nRF52840 microcontroller of the Arduino Stella)

The Arduino Stella redefines location tracking with its advanced microcontroller, the nRF52840 from Nordic Semiconductor, and the DCU040 Ultra-Wide Band (UWB) module from Truesense. Tailored for modern tracking needs, the Arduino Stella excels in pinpointing warehouse assets, ensuring healthcare safety and automating smart buildings.

The Arduino Stella

Seamlessly integrating with the Portenta UWB Shield and UWB-enabled smartphones through the dedicated NXP® Trimension™ App, Apple's Nearby Interaction APIs or Android's UWB Jetpack library, the Arduino Stella delivers robust finder functionality, precise point-to-point triggering and comprehensive tracking capabilities for applications demanding reliable, real-time location data.

Understanding UWB Technology

Ultra-Wideband (UWB) is a radio technology that uses very low energy levels for short-range, high-bandwidth communications over a large portion of the radio spectrum. Unlike traditional narrowband radio systems like Bluetooth® or Wi-Fi®, which operate in specific frequency bands, UWB transmits across a wide range of frequency bands (typically 500 MHz or more), allowing for greater precision in location tracking and higher data throughput.

UWB vs. Traditional Narrowband Technologies

The fundamental difference between UWB and traditional wireless technologies (Wi-Fi®, Bluetooth®, Zigbee® and Cellular) lies in their transmission methods:

Feature	Traditional Narrowband Radio	Ultra-Wideband Impulse Radio
Transmit Power	Higher transmit power	Lower transmit power
Initialization	Slow startup/initialization	Fast startup/initialization

Feature	Traditional Narrowband Radio	Ultra-Wideband Impulse Radio
Ranging	Poor ranging (signal strength based)	Excellent ranging (time of flight)
Multipath	Poor multipath robustness	Very good multipath robustness

Traditional narrowband systems use frequency or amplitude modulation to send data, requiring a reference carrier frequency between the transmitter and the receiver. This slows down the initialization process and restricts data transmission speed. UWB, in contrast, uses impulse radio technology. This technology consists of short pulses of energy (typically less than 5 nanoseconds) spread across a wide frequency band, allowing quick link initialization and extremely fast data transmission.

Key Characteristics of UWB

- **High precision**: UWB can determine the relative position of devices with centimeter-level accuracy (typically 5-10 cm), far more precise than GPS (meters), Bluetooth® (1-3 meters) or Wi-Fi® (2-15 meters).
- **Low-power consumption**: Despite its high data rates, UWB consumes very little power, making it suitable for battery-operated devices like the Arduino Stella, which is optimized for energy efficiency.
- **Short range**: Typically effective within 10-30 meters, making it ideal for indoor positioning applications where GPS signals are weak or unavailable.
- **Strong security**: The unique physical layer characteristics of UWB, including its wide bandwidth and low power spectral density, make it more resistant to jamming, eavesdropping, and relay attacks compared to other wireless technologies.
- **Immunity to multipath fading**: The wide bandwidth of UWB signals makes them highly resistant to multipath interference, where signals bounce off surfaces and create echoes.
- **Coexistence with other standards**: Due to its low output spectral power, UWB can operate alongside other wireless technologies without causing interference—other radio systems interpret UWB signals simply as background noise.

Spectrum and Frequencies

UWB operates in specific frequency bands regulated by telecommunications authorities worldwide. The Arduino Stella's UWB module (DCU040) uses channels in the 6.0 to 8.5 GHz range, allowing for high bandwidth communication while minimizing interference with other wireless technologies. In general, UWB can operate in the 3.1-10.6 GHz range, with channel widths of 500 MHz or greater.

How UWB Technology Works

UWB primarily uses a technique called "Time of Flight" (ToF) or "Time Difference of Arrival" (TDoA) to determine the distance between UWB-enabled devices with extraordinary precision. In a nutshell, this technique consists of the following steps:

1. An UWB transmitter sends a signal with a precise timestamp.
2. An UWB receiver detects the signal and calculates the time it took to arrive.
3. Since radio waves travel at the speed of light (approximately 30 cm per nanosecond), the system can calculate the distance with high precision.

For even greater accuracy, UWB can use the "Two-Way Ranging" (TWR) technique, where devices exchange several messages to account for clock synchronization issues. In a nutshell, this technique consists of the following steps:

1. Device A sends a message to device B and records the time (T1).
2. Device B receives the message and sends a response after a known delay.
3. Device A receives the response and records the time (T2).
4. The round-trip time, minus processing delays, determines the distance.

The Arduino Stella is typically configured as a Controller/Initiator in Two-Way Ranging scenarios, initiating the ranging process with other UWB devices like the Portenta UWB Shield. This mobile tag functionality makes it perfect for applications where tracking the position of moving objects is required.

Low Latency Advantage

The impulse radio technology of UWB provides an inherent advantage in terms of latency. The data

must modulate their carrier signals more slowly to maintain their frequency allocation, resulting in longer data latency and startup times.

In UWB systems, there's no need to establish a stable carrier signal for the receiver to lock onto, allowing data transmission to start immediately, representing a significant advantage for time-sensitive applications.

UWB System Components

In an UWB positioning system, devices typically operate in one of two roles:

- **Anchors**: Fixed-position devices (like the Portenta UWB Shield with a Portenta C33) that provide reference points for the positioning system. A minimum of three anchors is typically needed for 2D positioning and four for 3D positioning.
- **Tags**: Mobile devices that communicate with anchors to determine their position in space. The Arduino Stella is designed to function as a tag, with its compact form factor, battery operation and optimized power management, making it ideal for this role. Tags can also be integrated into smartphones, wearables or other IoT devices.

Applications of Arduino Stella with UWB Technology

The Arduino Stella's combination of UWB technology, compact form factor, and battery operation makes it ideal for numerous applications:

- **Asset tracking**: Attach Stella tags to valuable items in warehouses, hospitals, or construction sites to monitor their location with centimeter-level accuracy.
- **Wearable positioning**: Incorporate Arduino Stella into wearable devices for personnel tracking in industrial environments, healthcare settings, or event venues.
- **Robot and drone navigation**: Enable precise indoor navigation for autonomous systems where GPS is unavailable.
- **Social distancing solutions**: Create personal tags that can alert wearers when they come too close to others.
- **Interactive installations**: Power location-aware exhibits in museums or public spaces that respond to visitors' precise positions.
- **Sports performance analysis**: Track athletes' movements with high precision for training

Security applications: Create authorized personnel tags that can trigger secure access systems only when physically present (resistant to relay attacks).

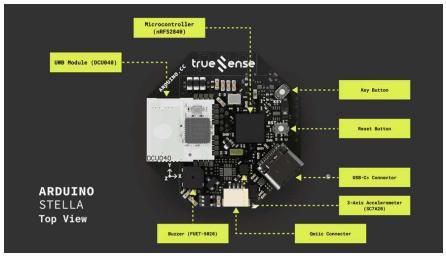
IoT sensor networks: Deploy Arduino Stella devices as mobile sensor nodes that can report both environmental data and precise location information.

The Arduino Stella's programmability, combined with its UWB capabilities, accelerometer, and Bluetooth® connectivity, makes it an extremely versatile platform for developing innovative positioning solutions.

Arduino Stella Architecture Overview

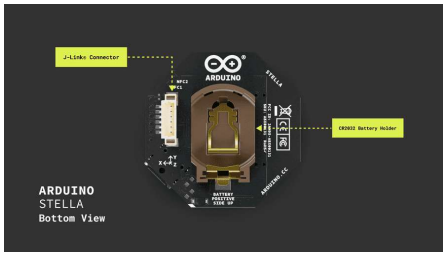
The Arduino Stella features a secure, certified and durable design that suits various applications, such as asset tracking, healthcare monitoring, smart building automation and consumer applications.

The top view of the Arduino Stella is shown in the image below:



The Arduino Stella's main components (top view)

The bottom view of the Arduino Stella is shown in the image below:



The Arduino Stella's main components (bottom view)

Here's an overview of the board's main components shown in the images:

Microcontroller: At the heart of the Arduino

Arm® Cortex®-M4 processor runs at 64 MHz and features 1 MB Flash and 256 kB RAM, providing ample processing power and memory for sophisticated applications.

UWB Module: The Truesense DCU040 module (based on the NXP® Trimension SR040 UWB IC) enables precise distance measurement with accuracy better than ±10 cm using two-way ranging techniques.

Connectivity: Besides UWB, the Arduino Stella includes Bluetooth® 5.0 connectivity through the nRF52840's integrated radio, supporting IEEE 802.15.4-2006 and 2.4 GHz transceiver capabilities.

Accelerometer: A 3-axis MEMS digital output accelerometer (SC7A20) enhances the board's motion detection capabilities, with programmable sensitivity ranges of ±2G/±4G/±8G/±16G and features like orientation detection, free-fall detection, and click detection.

User interfaces: The board includes a user-programmable button, a user-programmable LED and a buzzer (FUET-5020) that can be used for audible alerts.

Connectors: The Arduino Stella features a USB-C port for power and data, a QWIIC connector for I²C expansion, a CR2032 battery holder and a J-Link connector for debugging and alternative power input.

Board Libraries

The Arduino Stella and Portenta UWB Shield use different libraries and board cores due to their different microcontrollers and onboard UWB modules:

Arduino Stella Library

The `StellaUWB` library contains an application programming interface (API) to read data from the Arduino Stella and control its parameters and behavior. This library is designed to work with the DCU040 module on the Arduino Stella and supports the following:

One-way ranging (Time Difference of Arrival - TDoA) and two-way ranging (TWR).

Power management for battery-efficient operation.

Accelerometer control for motion detection.

Bluetooth® Low Energy connectivity for configuration and communication.

The Arduino Mbed OS Stella Boards core is required to work with the Arduino Stella's nRF52840 microcontroller.

Portenta UWB Shield Library

If you plan to use the Arduino Stella with a Portenta UWB Shield for two-way ranging, you'll also need the `PortentaUWBShield` library for the Portenta UWB Shield. This library is specifically designed for the DCU150 module used in the shield.

The Arduino Renesas Portenta Boards core is required to work with the Portenta C33 board that hosts the UWB Shield.

Bluetooth® Communication

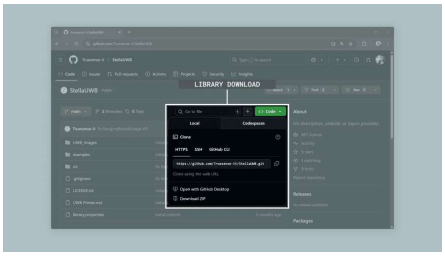
For examples that use Bluetooth® Low Energy communication, you'll also need the `ArduinoBLE` library. This library enables Bluetooth® Low Energy functionality for device discovery and initial connection setup before UWB ranging begins.

Installing the Libraries

To install the required libraries, follow these steps:

Step 1: Download the library ZIP file

- Visit the library's GitHub repository using the links provided above
- Click the green Code button
- Select Download ZIP from the dropdown menu
- Save the ZIP file to your computer



Library repository on GitHub

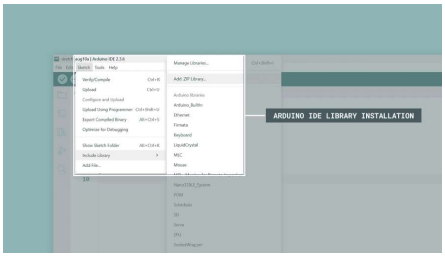
Step 2: Install the library in Arduino IDE

- Open the Arduino IDE
- Navigate to `Sketch > Include Library > Add .ZIP`

Browse to the location where you saved the ZIP file

Select the ZIP file and click Open

The library will be automatically installed and ready to use



Library installation on the Arduino IDE

Step 3: Verify the library installation

Go to `Sketch > Include Library`

Scroll down to see if the library appears in the `Contributed libraries` list

You can also check `File > Examples` to see if example sketches from the library are available

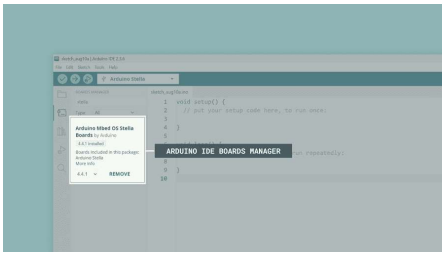
Installing the Board Cores

To install the required board cores:

Navigate to Tools > Board > Boards Manager...

For the Arduino Stella, search for `Arduino Mbed OS Stella Boards` and install the latest version

For the Portenta C33 board, search for `Arduino Renesas Portenta Boards` and install the latest version



Boards manager on the Arduino IDE

Important note: Make sure to install both the appropriate library and board core for your specific hardware. The Arduino Stella requires the `StellaUWB` library and `Arduino Mbed OS Stella Boards` core, while the Portenta UWB Shield with the Portenta C33 board requires the `PortentaUWBShield`

Pinout

The full pinout is available and downloadable as PDF from the link below:

Arduino Stella pinout

Datasheet

The complete datasheet is available and downloadable as PDF from the link below:

Arduino Stella datasheet

Schematics

The complete schematics are available and downloadable as PDF from the link below:

Arduino Stella schematics

STEP Files

The complete STEP files are available and downloadable from the link below:

Arduino Stella STEP files

First Use

Unboxing the Product

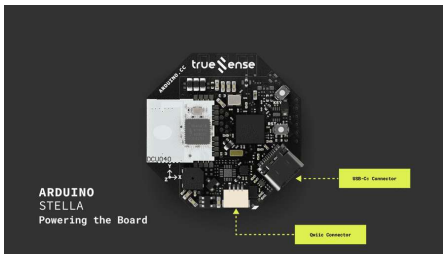
When you open the box of the Arduino Stella, you will find the board itself with its distinctive octagonal shape, featuring a USB-C port on one edge. The Arduino Stella is a compact board with dimensions of 38 mm x 38 mm, designed to be easily integrated into various tracking applications.

The Arduino Stella comes ready to use, but you may want to add a CR2032 battery if you plan to use it in portable applications that require battery power.

Powering the Board

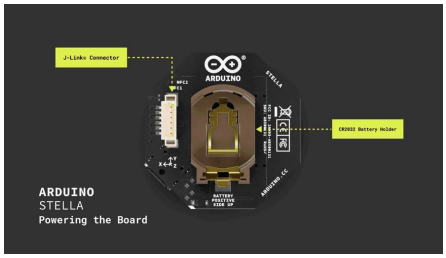
The Arduino Stella can be powered through one of these interfaces:

The USB-C port is the simplest and most common method, providing a stable +5 VDC power source when connected to a computer or wall adapter. The Qwiic connector can also supply power when connecting the board to a system that provides power through the Qwiic bus, enabling integration with other Qwiic-compatible components.



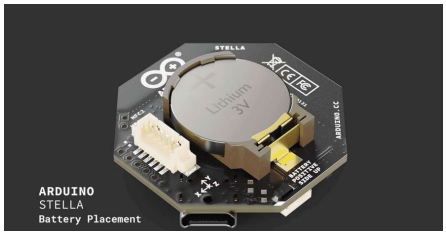
Powering options for the Arduino Stella (stationary applications)

For portable and wireless applications, the Arduino Stella offers two additional power options: the onboard CR2032 battery holder. The CR2032 battery holder allows you to install a +3 VDC coin cell battery, making the board completely wireless for mobile projects. The J-Link connector, while primarily used for debugging purposes, can also serve as a power input when needed.



Powering options for the Arduino Stella (portable and wireles applications)

Important note: When using battery power, ensure the correct polarity when inserting the CR2032 battery into the holder. The positive (+) side should face up, away from the PCB, as shown in the image below.



Correct battery placement in the Arduino Stella (positive side facing up)

Warning: When using Arduino Stella with USB-C power, remove the CR2032 battery; never power the Arduino Stella from both battery and USB simultaneously. The board is designed to prioritize USB power when connected. Removing the battery when working with USB will extend its life, ensure the most reliable operation and maximize battery lifespan.

Connecting to Your Computer

To program the Arduino Stella, connect it to your computer using a USB-C cable:

1. Insert the USB-C connector into the port on the Arduino Stella.
2. Connect the other end to an available USB port on your computer.

Once connected, you should see a power indicator light up on the board, indicating it's receiving power from the USB port.

Nearby World Example

Let's use the Arduino Stella to create a real-time distance measurement system using UWB technology. We will implement what we call the `Nearby World` example, which serves as our `Hello World` sketch for UWB technology. This example will verify the Arduino Stella's UWB capabilities and its ability to communicate with UWB-enabled smartphones.

This example sketch leverages Apple's Nearby Interaction protocol and similar UWB implementations on Android devices to establish a communication channel between the Arduino Stella and a UWB-enabled smartphone, allowing precise distance and angle measurements.

How It Works

The `Nearby World` example demonstrates the core functionality of UWB technology through a simple example sketch that can be described in the following key steps:

1. **Bluetooth® Low Energy connection setup**: The Arduino Stella broadcasts using Bluetooth® Low Energy to make itself discoverable to compatible smartphone apps.
2. **Configuration exchange**: The Bluetooth® Low Energy connection is used to exchange necessary UWB configuration parameters between the Arduino Stella and the smartphone.
3. **UWB ranging**: Once configured, the actual UWB ranging session begins, providing precise distance measurements.
4. **Real-time feedback**: Distance data is continuously updated and can be viewed both on the IDE's Serial Monitor and on the smartphone app.

This process demonstrates the working principle of many UWB applications, where Bluetooth® Low Energy is used primarily for discovery and configuration, while UWB handles the precise ranging.

Uploading the Sketch

Step 1: Verify Board Selection

Before uploading any code to the Arduino Stella, ensure you have selected the correct board:

1. Connect the Arduino Stella to your computer using a USB-C cable
2. Open the Arduino IDE
3. **Critical:** Navigate to
   ```
   Tools > Board > Arduino Mbed OS Stella Boards > Arduino Stella
   ```
4. Verify the correct port is selected in `Tools > Port`

Important note: Do not select `Arduino BLE Sense 33`, `Portenta C33` or any other board when programming the Arduino Stella. This will cause compilation errors.

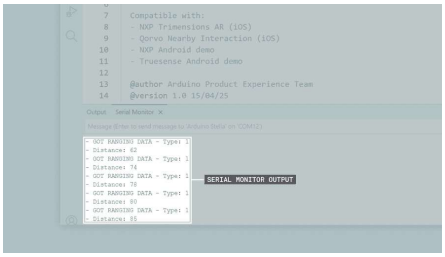
Step 2: Upload the Sketch

Copy and paste the example sketch below into a new sketch in the Arduino IDE:

```
/**
  Nearby World Example for Arduino
  Name: arduino_stella_nearby_world
  Purpose: This sketch demonstrates
  to measure distance between the b

  Compatible with:
  - NXP Trimensions AR (iOS)
  - Qorvo Nearby Interaction (iOS)
  - NXP Android demo
  - Truesense Android demo

  @author Arduino Product Experienc
  @version 1.0 15/04/25
*/

// Include required libraries
#include <ArduinoBLE.h>
#include <StellaUWB.h>

// Track the number of connected Bl
uint16_t numConnected = 0;

/**
  Processes ranging data received f
  @param rangingData Reference to u
*/
void rangingHandler(UWBRangingData
```

To upload the sketch to the Arduino Stella, click the **Verify** button to compile the sketch and check for errors, then click the **Upload** button to program the device with the example sketch.

Once the sketch is uploaded, open the Serial Monitor by clicking on the icon in the top right corner of the Arduino IDE. You should see the message `- Nearby interaction app start...` in the IDE's Serial Monitor. When a smartphone connects, you'll see `Client connected!` followed by `Session started!` and then real-time distance measurements in millimeters. If no device connects, only the initialization message will be displayed.



Arduino Stella Serial Monitor showing initial connection and distance measurements

Try It Yourself

To complete the test, you will need a UWB-enabled smartphone with one of the compatible applications installed:

For iPhone (iPhone 11 or newer with UWB capability):

- NXP® Trimensions AR
- Qorvo Nearby Interaction

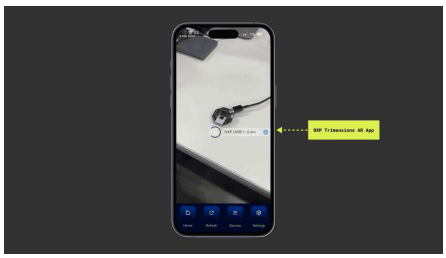
For Android (UWB-enabled Android devices):

- Truesense Android demo
- NXP® Android demo

> **Important note for Android devices:** If Developer Options is currently enabled on your Android device, please ensure all radio-related options remain at their default settings. We recommend keeping Developer Options disabled for optimal UWB functionality. If you previously enabled it, consider disabling it for the most stable device operation.

Install one of these apps on your smartphone and follow these steps:

1. Open the app on your smartphone
2. Look for a device named `Arduino Stella` in the app's device list
3. Connect to the device
4. Once connected, the app will initiate a UWB ranging session
5. Move your phone closer to and further from the Arduino Stella



Smartphone app displaying UWB connection and real-time distance to Arduino Stella

The smartphone app shows the connection status and continuously updates the distance measurement as you move the phone. Both the app

and the Serial Monitor display synchronized measurements in millimeters.

You should see the distance measurements updating in real-time both on your smartphone app and in the IDE's Serial Monitor. The distances are shown in millimeters, providing centimeter-level accuracy characteristic of UWB technology.

NearbyDemo Example

About the NearbyDemo Example

Before you begin: Ensure you have selected `Tools > Board > Arduino Mbed OS Stella Boards > Arduino Stella` in the Arduino IDE. Selecting the wrong board (like `Arduino BLE Sense 33` or the `Portenta C33`) will prevent this code from compiling.

The `NearbyDemo` example sketch demonstrates how to implement distance measurement between the Arduino Stella and UWB-enabled smartphones using Apple's Nearby Interaction protocol or compatible Android implementations. This example showcases how Bluetooth® Low Energy is used for initial device discovery and configuration exchange, while UWB handles the precise distance measurements.

This example demonstrates the following:

- **Hybrid communication protocol**: Integration of Bluetooth® Low Energy for device discovery and configuration with UWB for precise distance measurements
- **Standards compatibility**: Compatible with Apple's Nearby Interaction API and Android UWB implementations
- **Power-efficient design**: UWB subsystem only activates when a client connects, conserving battery life
- **Multi-client support**: Can handle connections from multiple smartphones simultaneously

Real-world applications for this example include:

- **Asset tracking**: Precisely locate Arduino Stella devices attached to valuable items using smartphones
- **Smart building automation**: Implement room-level presence detection for environmental control

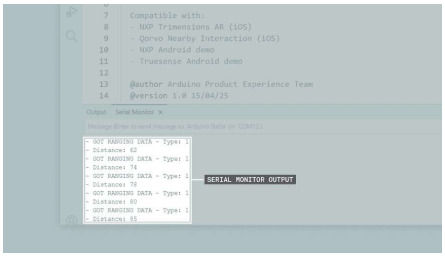
Retail analytics: Analyze customer movement patterns using their smartphones

Consumer item finding: Create smartphone-compatible tags for locating misplaced items

Here's the complete code for the `NearbyDemo` example sketch:

```
/**
  NearbyDemo Example for Arduino St
  Name: arduino_stella_nearbydemo.i
  Purpose: This sketch demonstrates
  to measure distance between the b

  Compatible with:
  - NXP Trimensions AR (iOS)
  - Qorvo Nearby Interaction (iOS)
  - NXP Android demo
  - Truesense Android demo

  @author Arduino Product Experienc
  @version 1.0 15/04/25
*/

// Include required libraries
#include <ArduinoBLE.h>
#include <StellaUWB.h>

// Track the number of connected Bl
uint16_t numConnected = 0;

/**
  Processes ranging data received f
  @param rangingData Reference to l
*/
void rangingHandler(UWBRangingData
```

Once the example sketch is uploaded and a smartphone connects to the Arduino Stella, the IDE's Serial Monitor will display the connection and ranging information:



Arduino Stella Serial Monitor output showing Nearby Demo connection and distance measurements

smartphone establishes a Bluetooth® Low Energy connection. Only after the connection is established will you see session status and real-time distance measurements in millimeters. If no device connects, you will only see the initial "Nearby interaction app start..." message waiting for a connection.

> **Important note**: Distance measurements only appear **after** a successful connection with a UWB-enabled smartphone. Without a connection, the Serial Monitor will only show the initialization message.

Key Components of the Example Sketch

The `NearbyDemo` code implements an event-driven architecture using callback functions to handle various stages of the UWB communication process. Let's examine the main components:

Libraries and Global Variables

```
#include <ArduinoBLE.h>
#include <StellaUWB.h>

uint16_t numConnected = 0;
```

The example sketch uses two important libraries for its operation:

- `ArduinoBLE`: Provides Bluetooth® Low Energy functionality for device discovery and initial connection.
- `StellaUWB`: The core library that enables interaction with the onboard UWB module of the Arduino Stella.

The `numConnected` variable tracks how many Bluetooth® Low Energy clients are currently connected to the Arduino Stella.

Ranging Data Handler

The heart of the UWB functionality resides in the ranging callback:

```
void rangingHandler(UWBRangingData &ra
  if(rangingData.measureType() == (uin
    RangingMeasures twr = rangingData.
    // Process measurements...
  }
}
```

This function processes incoming distance measurements, validates the data integrity, and outputs results to the IDE's Serial Monitor. The validation checks ensure only valid measurements are displayed (`status = 0` and `distance ≠ 0xFFFF`).

Connection Management

The connection callbacks manage the UWB subsystem lifecycle efficiently:

```
void clientConnected(BLEDevice dev) {
  if (numConnected == 0) {
    UWB.begin();
  }
  numConnected++;
}
```

This power-saving approach ensures the UWB hardware only activates when needed, extending battery life in portable applications.

Session Management

```
void sessionStarted(BLEDevice dev) {
  Serial.println("- Session started!")
}
```

These functions provide feedback about the session lifecycle, helping developers understand the connection state during debugging.

Setup and Initialization

The setup function registers all callbacks and starts Bluetooth® Low Energy advertising:

```
void setup() {
  // Register all callbacks
  UWB.registerRangingCallback(rangingH
  UWBNearbySessionManager.onConnect(cl
  // ... other callbacks

  // Start advertising with device nam
  UWBNearbySessionManager.begin("Ardui
}
```

The device name "Arduino Stella" appears in smartphone apps when scanning for available UWB devices.

Main Loop

The main loop maintains the Bluetooth® Low Energy connection:

```
void loop() {
  delay(100);
  UWBNearbySessionManager.poll();
}
```

The `poll()` function processes Bluetooth® Low Energy events while the actual UWB ranging occurs asynchronously through callbacks.

Testing with Smartphones

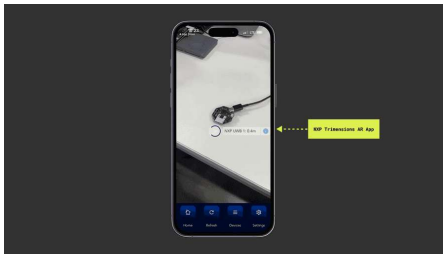
To test this example with a UWB-enabled smartphone:

1. Upload the sketch to your Arduino Stella
2. Open the IDE's Serial Monitor at 115200 baud
3. Install a compatible app on your smartphone:

   For iOS: NXP® Trimensions AR or Qorvo Nearby Interaction

   For Android: Truesense or NXP® UWB demo apps

1. Connect to the device named "Arduino Stella" in the app
2. Move your phone to see distance measurements update in real-time



Smartphone app showing UWB connection and distance measurement to Arduino Stella

The smartphone app displays the connection status and real-time distance to the Arduino Stella. Once connected, both the app and the IDE's Serial Monitor will show synchronized distance measurements as you move the phone. The distance measurements are displayed in millimeters, providing centimeter-level accuracy. For example, a reading of "Distance: 80" indicates approximately 8 cm between devices.

> **Important note**: If the connection fails or no measurements appear, verify that your smartphone has UWB enabled and that the Arduino Stella is powered on and running the sketch.

Extending the Example Sketch

The `NearbyDemo` example sketch provides a great foundation that you can build upon for more complex projects. Some possible extensions of this example sketch include the following:

- **Visual feedback**: Add LED patterns based on distance thresholds
- **Audio alerts**: Implement buzzer feedback for proximity warnings
- **Motion detection**: Combine accelerometer data with ranging for activity monitoring
- **Data logging**: Record distance measurements for analysis
- **Custom device names**: Modify the advertising name for multiple device deployments

The event-driven architecture makes it easy to add features without disrupting the core ranging functionality.

Two-Way Ranging Example

📋 **Hardware Setup Reminder:**

For the Arduino Stella: Select `Tools > Board > Arduino Mbed OS Stella Boards > Arduino Stella` on the Arduino IDE

For the Portenta C33 board and the UWB Shield: Select `Tools > Board > Arduino Renesas Portenta Boards > Portenta C33` on the Arduino IDE

The Two-Way Ranging example demonstrates direct UWB communication between two Arduino devices: the Arduino Stella (acting as a Controller/Initiator) and the Portenta UWB Shield (acting as a Controlee/Responder). This example showcases the fundamental distance measurement capabilities of UWB technology in a dedicated device-to-device setup without requiring external UWB-enabled consumer devices such as smartphones.

> **Important note**: In UWB communication, the terms "Controller" and "Controlee" refer to specific roles within a ranging session. A **Controller** (also called an Initiator) is the device that initiates and controls the ranging process, sending the initial signal and managing the timing of exchanges. A **Controlee** (also called a Responder) is the device that responds to the Controller's signals. These terms are used interchangeably in UWB documentation: Controller/Initiator and Controlee/Responder refer to the same roles. In positioning systems, Controllers/Initiators often correspond to mobile "tags" while Controlees/Responders often serve as stationary "anchors".

This example demonstrates the following:

Direct device-to-device communication: Unlike the `NearbyDemo` example, which requires a smartphone, this example establishes direct UWB communication between two UWB-enabled Arduino devices.

Controller-Controlee architecture: It shows how to configure one device as a Controller (initiator of the ranging) and another as a Controlee (responder).

Double-Sided Two-Way Ranging (DS-TWR): This technique provides higher accuracy in distance measurements by accounting for clock drift between devices.

Visual feedback system: Both devices provide LED feedback to indicate connection status and distance measurements.

smoothed moving average for analysis.

Some of the real-life applications for this example include:

- **Multi-node positioning systems**: Creating networks of UWB nodes for advanced indoor positioning.
- **Robot navigation**: Enabling precise distance measurements between robots or between robots and fixed stations.
- **Asset tracking**: Building custom tracking solutions with multiple Arduino-based UWB anchors.
- **Proximity detection systems**: Creating safety systems that can detect precise distances between industrial equipment and personnel.
- **Access control systems**: Implementing secure entry systems based on precise proximity detection.
- **Interactive installations**: Enabling position-based interactive exhibits in museums or public spaces

Here's the code for the Arduino Stella, which acts as the Controller (Initiator) in this Two-Way Ranging scenario.

📋 **Board Selection Check:** Ensure `Arduino Stella` is selected in the Arduino IDE before uploading this code.

```
/**
  Two-Way Ranging Controller Exampl
  Name: stella_uwb_twr_controller.i
  Purpose: This sketch configures t
  for Two-Way Ranging with a Porter
  The LED provides visual feedback

  @author Arduino Product Experienc
  @version 1.0 15/04/25
*/

// Include required UWB library
#include <StellaUWB.h>

// Pin definitions
#define LED_PIN p37  // Stella's bu

// Distance and timing parameters
#define MAX_DISTANCE 300     // Max
#define MIN_BLINK_TIME 50    // Fas
#define MAX_BLINK_TIME 1000  // Slo
#define TIMEOUT_MS 2000      // Cor

// System state variables
unsigned long lastBlink = 0;
unsigned long lastMeasurement = 0;
bool ledState = false;
int currentBlinkInterval = MAX_BLIN
```

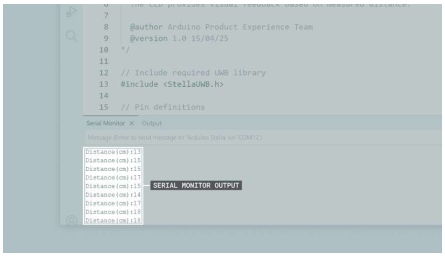
Here's the code for the Portenta UWB Shield, which acts as the Controlee (Responder) in this Two-Way Ranging scenario:

📋 **Board Selection Check:** Ensure `Portenta C33` is selected in the Arduino IDE before uploading this code.

```
/**
  Two-Way Ranging Controlee Example
  Name: portenta_uwb_twr_controlee.
  Purpose: This sketch configures t
  for Two-Way Ranging with an Ardui
  Includes distance visualization a

  @author Arduino Product Experienc
  @version 1.0 15/04/25
*/

// Include required UWB library
#include <PortentaUWBShield.h>

// Moving average configuration
#define SAMPLES 10
long distances[SAMPLES] = {0};
int sample_index = 0;

// LED and status configuration
#define NEARBY_THRESHOLD 300
#define CONNECTION_TIMEOUT 2000
#define LED_BLINK_INTERVAL 500

// System state variables
unsigned long lastMeasurement = 0;
unsigned long lastLedBlink = 0;
bool ledState = false;
```

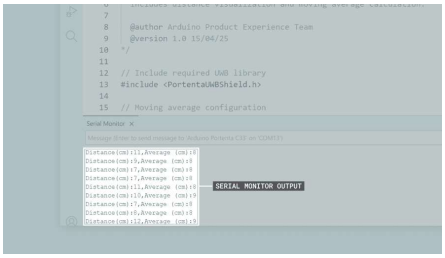
Important note: Both devices must be programmed and powered on before you will see any distance measurements. The ranging session only begins when both the Controller (Arduino Stella) and Controlee (Portenta UWB Shield) are running their respective sketches.

Once both devices are running, the Arduino Stella's Serial Monitor will display the distance measurements:



Arduino Stella Serial Monitor output showing distance measurements

Similarly, the Portenta UWB Shield's Serial Monitor will show both raw and averaged distance measurements:



Portenta UWB Shield Serial Monitor output showing distance and average measurements

The output format from the Portenta C33 board is optimized for visualization, showing both the immediate distance readings and the calculated moving average.

Key Components of the Example Sketch

The Two-Way Ranging example uses simplified helper classes (`UWBTracker` and `UWBRangingControlee`) that make device configuration easier while maintaining the same functionality. Let's analyze the key components:

Libraries and MAC Addressing

Both devices use their respective UWB libraries:

- The Arduino Stella uses `StellaUWB.h` (for the DCU040 module)
- The Portenta UWB Shield uses `PortentaUWBShield.h` (for the DCU150 module)

MAC address configuration remains critical for communication:

```
// On Arduino Stella (Controller)
uint8_t devAddr[] = {0x22, 0x22};
uint8_t destination[] = {0x11, 0x11};

// On Portenta UWB Shield (Controlee)
uint8_t devAddr[] = {0x11, 0x11};
uint8_t destination[] = {0x22, 0x22};
```

Important note: The MAC addresses are reversed between the two devices. The Arduino Stella identifies itself as `0x2222` and expects to communicate with `0x1111`, while the Portenta UWB Shield identifies itself as `0x1111` and expects to communicate with `0x2222`. Both devices must use the same session ID (`0x11223344`) to establish communication.

Simplified Session Setup

The new code uses helper classes that simplify the UWB session configuration:

Arduino Stella (Controller/Initiator):

```
UWBTracker myTracker(0x11223344, srcAd
UWBSessionManager.addSession(myTracker
myTracker.init();
myTracker.start();
```

Portenta UWB Shield (Controlee/Responder):

```
UWBRangingControlee myControlee(0x1122
UWBSessionManager.addSession(myControl
myControlee.init();
myControlee.start();
```

These helper classes automatically configure the appropriate ranging parameters for their respective roles, making the setup process more straightforward.

Enhanced Visual Feedback

Both devices in this example provide visual feedback to help users understand the system status and distance measurements at a glance. Each device uses its LED capabilities to indicate different operational states.

Arduino Stella LED Behavior:

The Arduino Stella uses its built-in LED to provide distance-based feedback through variable blink rates:

- **Close range (0 to 50 cm)**: Very fast blinking for immediate proximity alert

- **Far range (150 to 300 cm)**: Slow blinking indicating increasing distance
- **Very far (>300 cm)**: Very slow blinking for maximum range
- **No connection**: Rapid blinking (100 ms intervals) as a warning signal
- **Startup**: Triple flash to indicate successful initialization

This variable blink rate creates an intuitive feedback system where users can gauge distance without looking at the serial output.

Portenta C33 RGB LED Behavior:

The Portenta C33 uses its RGB LED to provide feedback about different aspects of the system operation:

- **Red LED**: System heartbeat (continuous blinking shows the system is running)
- **Blue LED**: Connection status (ON = no connection, OFF = connected to tag)
- **Green LED**: Proximity indicator (ON = tag within 300 cm threshold, OFF = tag beyond threshold)

This three-color system provides immediate visual feedback about connection status, system operation, and proximity all at once, making it useful for debugging and demonstration purposes where checking serial output isn't practical.

Data Processing and Visualization

The Portenta UWB Shield implements a moving average filter to smooth distance measurements:

```
// Store measurements in circular buff
distances[sample_index] = twr[j].dista

// Calculate moving average
long avg = 0;
for (int i = 0; i < SAMPLES; i++) {
  avg += distances[i];
}
avg = avg / SAMPLES;
```

This provides two data streams for visualization:

- **Raw distance**: Immediate measurements showing real-time variations
- **Averaged distance**: Smoothed values for trend analysis

```
Serial.print("- Distance (cm):");
Serial.print(twr[j].distance);
Serial.print(",");
Serial.print("- Average (cm):");
Serial.println(avg);
```

Try It Yourself

To test this two-device ranging setup, you will need to prepare both the hardware connections and software configuration before running the example.

Hardware Setup:

First, prepare your devices by following these steps:

1. Connect the Portenta UWB Shield to your Portenta C33 board
2. Power the Arduino Stella (using a USB-C cable or a CR2032 battery)

Software Setup:

Next, upload the appropriate sketches to each device if it was not done before:

1. **For the Portenta C33 board and the Portent UWB Shield:**

   Select `Tools > Board > Arduino Renesas Portenta Boards > Portenta C33`

   Upload the Controlee/Responder sketch to the Portenta C33 board

1. **For the Arduino Stella:**

   Select `Tools > Board > Arduino Mbed OS Stella Boards > Arduino Stella`

   Upload the Controller/Initiator sketch to the Arduino Stella

Important note: If you encounter compilation errors, verify that you have selected the correct board for each device before uploading.

Testing:

Once both devices are programmed, you can begin testing the ranging system:

1. Open the IDE's Serial Monitor for the Portenta C33 board at 115200 baud
2. For best visualization, use `Tools > Serial Plotter` instead of the `Serial Monitor`
3. Move the devices closer and further apart
4. Observe the LED feedback on both devices

> If you don't see any measurements, verify that both devices are powered on and running their respective sketches. The system requires both the Controller and Controlee to be active for ranging to occur.

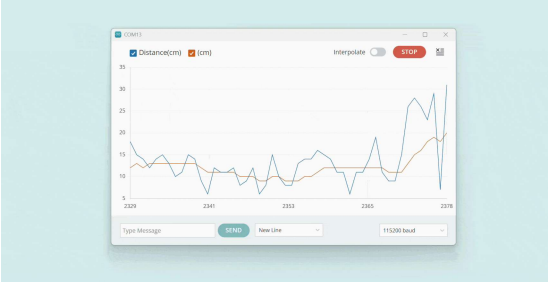
Visual Feedback During Operation:

When the system is working correctly, both devices provide LED feedback:

- **Arduino Stella LED:** Blinks faster as devices get closer (fast at 50 cm, slow at 300 cm)
- **Portenta C33 Red LED:** Blinks continuously to show system is active
- **Portenta C33 Blue LED:** OFF when connected, ON when connection is lost
- **Portenta C33 Green LED:** ON when Stella is within 300 cm, OFF when further away

Serial Output Visualization:

The Serial Plotter provides real-time visualization of the distance measurements:



The graph shows two lines:

- **Blue line:** Raw distance measurements in centimeters
- **Red line:** Smoothed moving average for trend analysis

Example readings:
`Distance(cm):125,Average(cm):123` indicates the devices are approximately 1.25 meters apart.

Extending the Two-Way Ranging Example

This enhanced example provides a foundation for more advanced applications:

- **Access control systems**: Use the distance threshold to trigger door locks or security systems
- **Safety zones**: Create multiple distance thresholds for industrial safety applications
- **Data logging**: Add SD card storage to record distance measurements over time
- **Multi-node networks**: Extend to support multiple tags communicating with one base station
- **Position calculation**: Combine multiple base stations for 2D/3D positioning through triangulation
- **Alert systems**: Add buzzer or additional LEDs for specific distance-based alerts

The visual feedback and data processing capabilities demonstrated in this example can be adapted to various real-world applications requiring precise distance measurement and proximity detection.

Support

If you encounter any issues or have questions while working with your Arduino Stella, we provide various support resources to help you find answers and solutions.

Help Center

Explore our Help Center, which offers a comprehensive collection of articles and guides for Arduino boards and shields. The Help Center is designed to provide in-depth technical assistance and help you make the most of your device.

Arduino Help Center

Forum

Join our community forum to connect with other Arduino Stella users, share your experiences, and ask questions. The Forum is an excellent place to learn from others, discuss issues, and discover new ideas and projects related to the Arduino Stella.

Arduino Stella in the Arduino Forum

Please get in touch with our support team if you need personalized assistance or have questions not covered by the help and support resources described before. We're happy to help you with any issues or inquiries about the Arduino Stella.

Contact us page

Suggest changes

The content on docs.arduino.cc is facilitated through a public GitHub repository. If you see anything wrong, you can edit this page here.

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